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(54) **CONNECTOR HAVING VIBRATION
ABSORPTION AND NOISE REMOVAL
PROPERTIES**

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H01R 13/15 (2006.01)
H01R 13/533 (2006.01)
H01R 24/40 (2011.01)

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CPC **H01R 13/719** (2013.01); **H01R 13/15**
(2013.01); **H01R 13/533** (2013.01); **H01R**
24/40 (2013.01)

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CPC H01R 13/719; H01R 13/15; H01R 13/533;
H01R 24/40; H01R 13/7197; H01R 24/42
See application file for complete search history.

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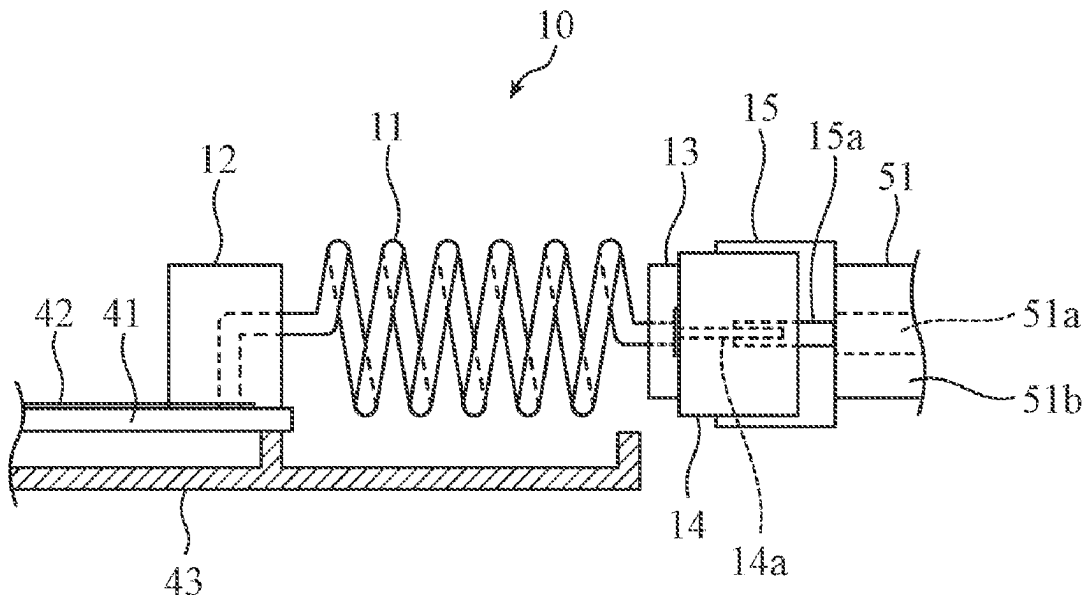
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(57) **ABSTRACT**

A connector includes a spring structure having a vibration
absorption property and an inductance component having
noise removal properties. One end of the spring structure is
configured to be connected to a wiring pattern provided on
a board via a support member, and an opposite end of the
spring structure is configured to be connected to an electric
wire via a connector.

4 Claims, 4 Drawing Sheets



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FIG. 1

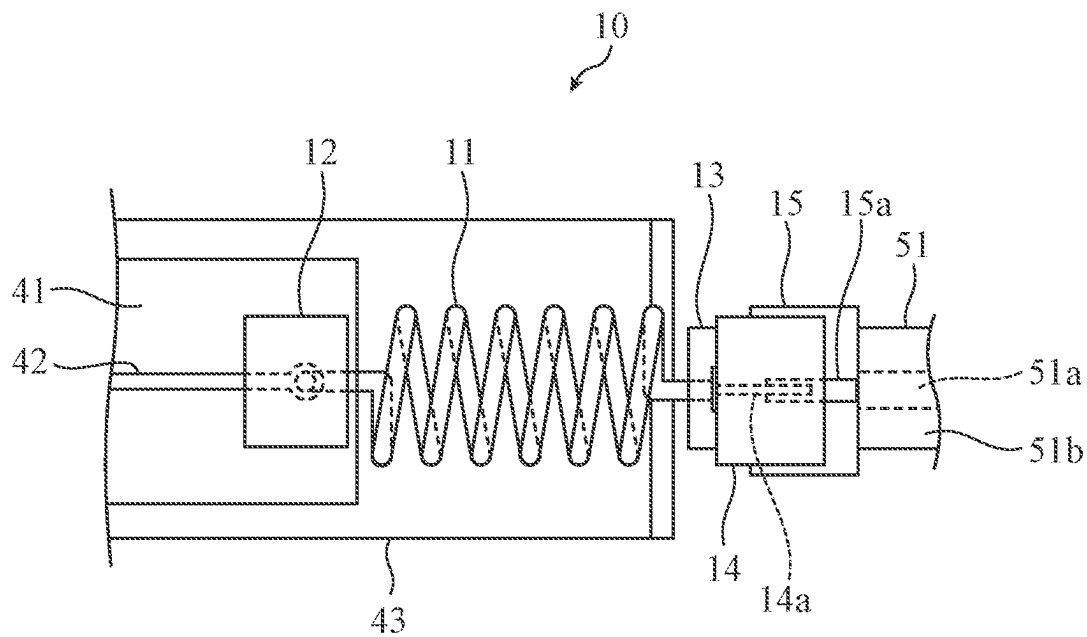


FIG. 2

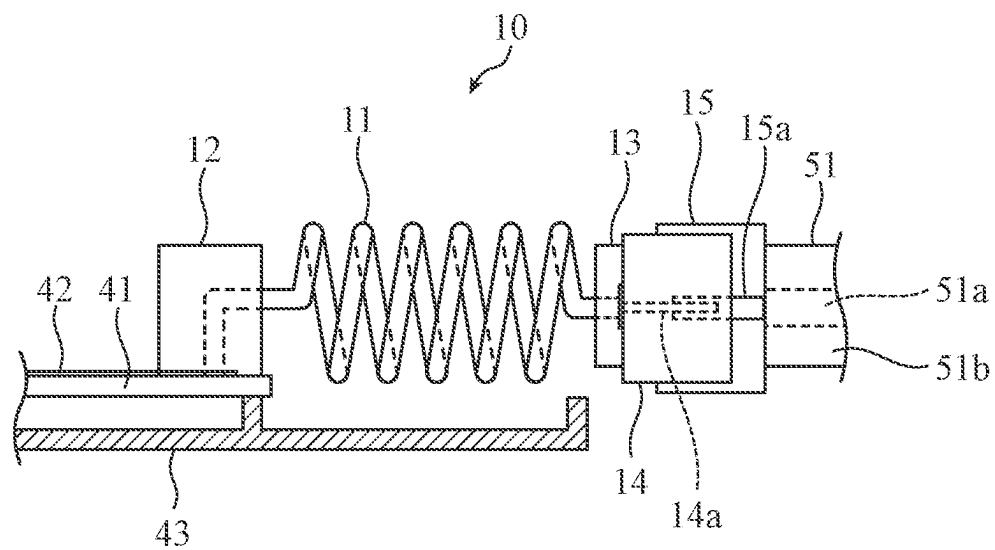


FIG. 3

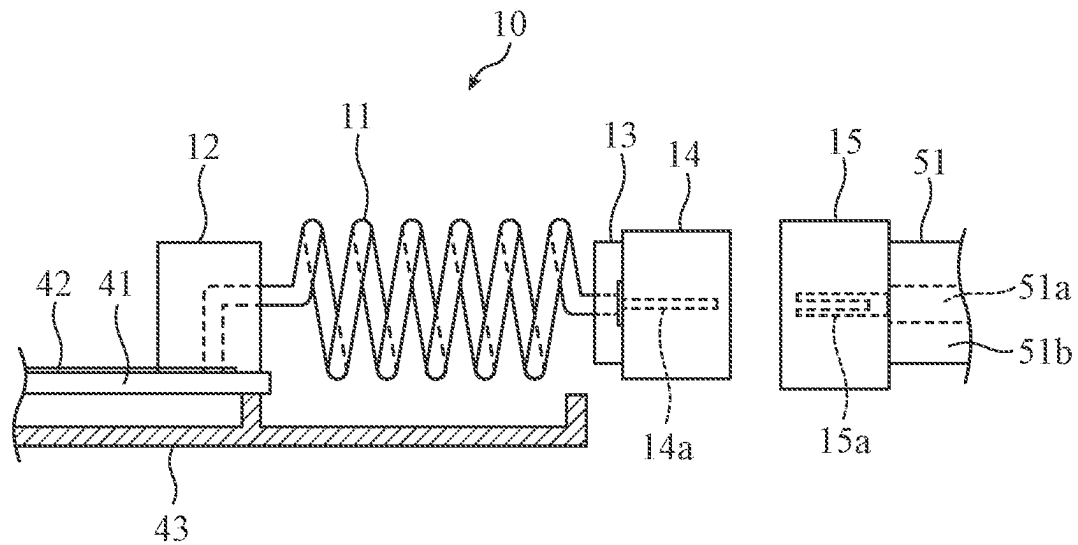


FIG. 4

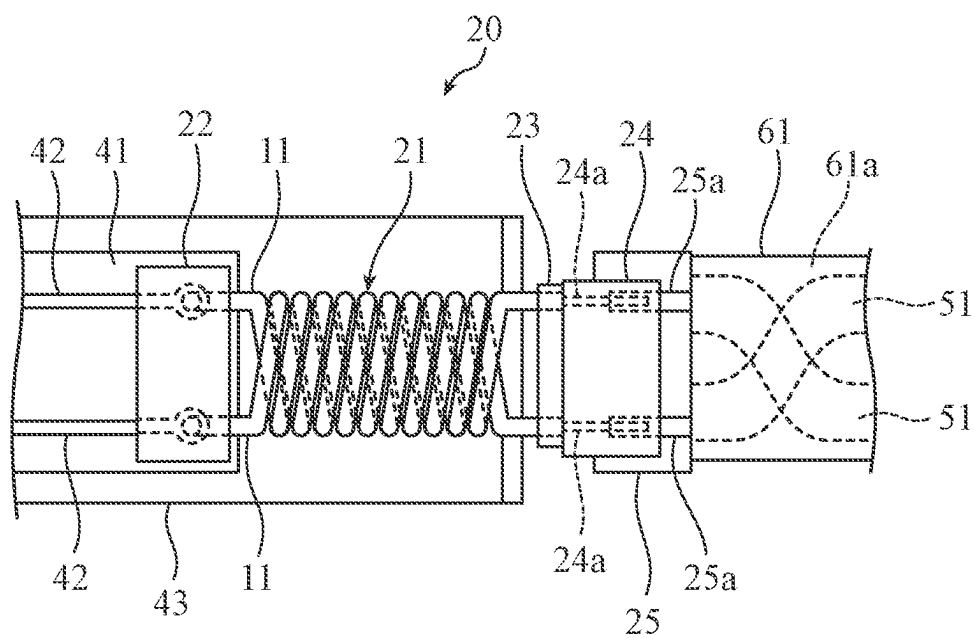


FIG. 5

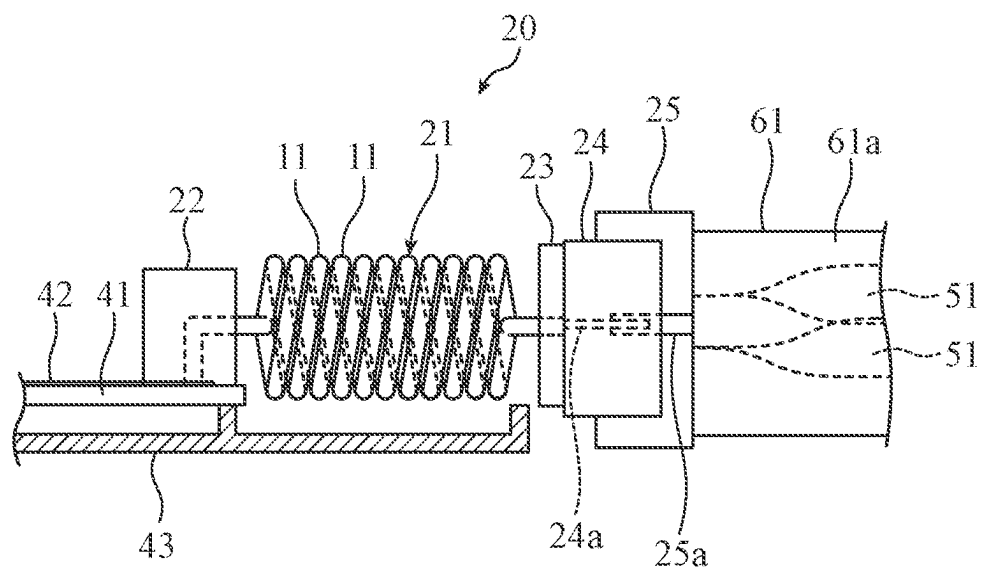


FIG. 6A

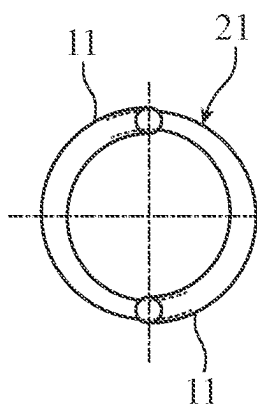


FIG. 6B

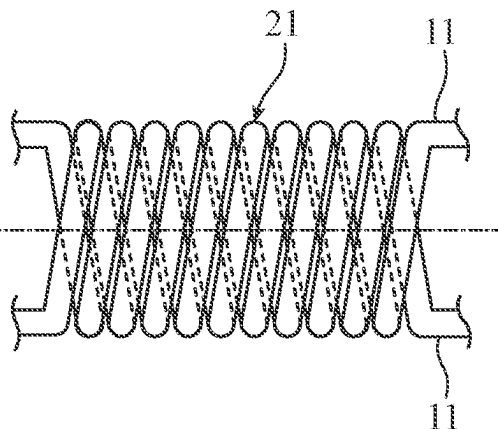
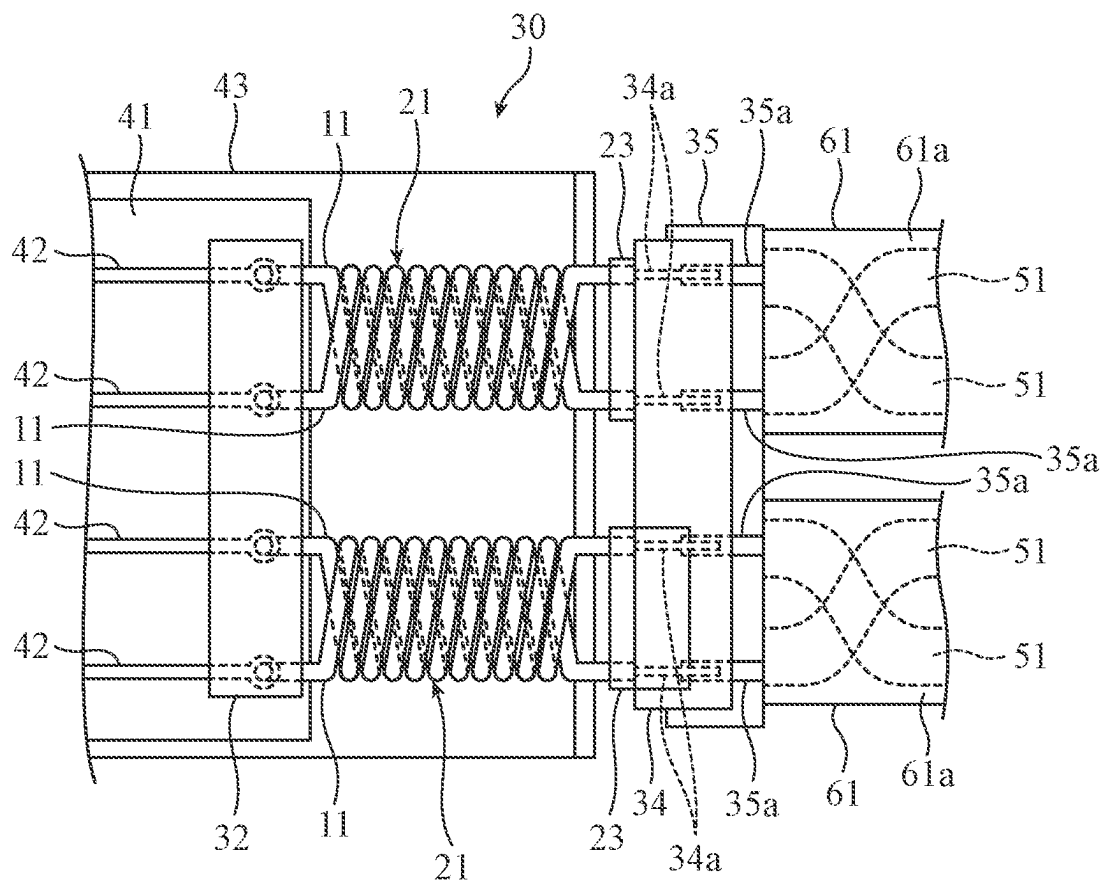


FIG. 7



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CONNECTOR HAVING VIBRATION ABSORPTION AND NOISE REMOVAL PROPERTIES

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of PCT International Application No. PCT/JP2020/025049, filed on Jun. 25, 2020, which is hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

The present disclosure relates to a connector.

BACKGROUND ART

Connectors are parts for connecting a flow of power or electric signals. Such a connector electrically connects an electric wire and electronic equipment, for example. Therefore, there is a case in which an electrical noise is transmitted from the electric wire to the electronic equipment via the connector. In this case, the electronic equipment to which the noise is propagated has a possibility of malfunctioning.

In Patent Literature 1, a technique of reducing noises propagated to electronic equipment is described. The technique disclosed in Patent Literature 1 reduces a noise propagated to electronic equipment by providing a coiled conducting wire between the electric equipment and a coaxial cable.

CITATION LIST

Patent Literature

Patent Literature 1: Japanese Unexamined Patent Application Publication No. JP S 53-78701 A

SUMMARY OF INVENTION

Technical Problem

A connector electrically connects between an electric wire and electronic equipment, and also connects between them mechanically. Therefore, there is a case in which a mechanical vibration is transmitted either from the electric wire to the electronic equipment or from the electronic equipment to the electric wire. In this case, there is a possibility that either the electronic equipment or the connector gets damaged. It is impossible for the technique disclosed in Patent Literature 1 to absorb mechanical vibrations.

The present disclosure is made in order to solve the above-mentioned problems, and it is therefore an object of the present disclosure to provide a connector that can achieve both the absorption of vibrations and the removal of noises using a single spring structure.

Solution to Problem

A connector according to the present disclosure includes: a spring structure having a spring property as a result of being constituted by a conducting wire wound spirally, and functioning as a coil having an inductance component; a one end side support member to support one end of the spring structure, and to connect the one end of the spring structure to a wiring pattern provided on a board; and an other end

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side support member to support the other end of the spring structure, and to electrically connect the other end of the spring structure to an electric wire, in which the spring structure is constituted by a pair of conducting wires wound spirally and alternately each other, one ends of the pair of conducting wires are connected to a pair of respective wiring patterns provided on the board, the other ends of the pair of conducting wires are connected to a pair of respective electric wires contained in a single cable, the pair of conducting wires are provided coaxially and in a same winding direction, and an insulator is provided on each of outer peripheral surfaces of the pair of conducting wires.

Advantageous Effects of Invention

According to the present disclosure, both the absorption of vibrations and the removal of noises can be achieved using a single spring structure.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a plane view showing the configuration of a connector according to Embodiment 1;

FIG. 2 is a side view showing the configuration of the connector according to Embodiment 1;

FIG. 3 is a view showing a state in which the connector according to Embodiment 1 and an electric wire are not connected mechanically and electrically;

FIG. 4 is a plane view showing the configuration of a connector according to Embodiment 2;

FIG. 5 is a side view showing the configuration of the connector according to Embodiment 2;

FIG. 6A is an outline view showing a double spring structure which is viewed from an axis direction thereof, and

FIG. 6B is an outline view showing the double spring structure which is viewed from the outside in a radial direction thereof; and

FIG. 7 is a plane view showing the configuration of a connector according to Embodiment 3.

DESCRIPTION OF EMBODIMENTS

Hereinafter, in order to explain the present disclosure in greater detail, embodiments of the present disclosure will be explained with reference to the accompanying drawings.

Embodiment 1

A connector 10 according to Embodiment 1 will be explained using FIGS. 1 to 3.

FIG. 1 is a plane view showing the configuration of the connector 10 according to Embodiment 1. FIG. 2 is a side view showing the configuration of the connector 10 according to Embodiment 1. FIG. 3 is a view showing a state in which the connector 10 according to Embodiment 1 and an electric wire 51 are not connected mechanically and electrically.

As shown in FIGS. 1 and 2, the connector 10 according to Embodiment 1 mechanically and electrically connects a single board 41 and the single electric wire 51, for example. One end of the connector 10 is mechanically and electrically connected to the board 41, and the other end of the connector 10 is mechanically and electrically connected to the electric wire 51.

The board 41 has a single wiring pattern 42. This wiring pattern 42 is formed of, for example, copper or the like, and is provided on an upper surface of the board 41. Further, the

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board **41** is fixed to a housing **43** of electronic equipment (not illustrated). More specifically, the board **41** is intended for controlling the electronic equipment.

The electric wire **51** is the one in which a signal line **51a** serving as a conductor is covered by an insulator **51b**. To one end of this electric wire **51** is provided a socket **15** which will be mentioned later.

The connector **10** includes a spring structure **11**, supporting members **12** and **13**, a plug **14** and the socket **15**.

The spring structure **11** is supported between the support members **12** and **13**. This spring structure **11** has a function of absorbing mechanical vibrations and a function of removing electrical noises.

Concretely, the spring structure **11** is the one in which a single conducting wire is spirally wound. Therefore, the spring structure **11** has a spring property. More specifically, the spring structure **11** can be elastically deformed in an axis direction thereof. The spring structure **11** also functions as a coil having an inductance component.

The conducting wire has, for example, a circular cross section, but its cross section may have an elliptic shape, a rectangular shape or another shape. Further, the spring structure **11** is constituted by a conducting wire wound spirally, and, as a result, its cross section is a circle. In addition, as for the conducting wire which constitutes the spring structure **11**, a metal having high tensile strength and high electrical conductivity is suitable. The conducting wire is formed of, for example, at least one metallic material out of aluminum, copper, nickel, silver, etc.

The supporting member **12** constitutes a one end side support member. This supporting member **12** is formed of, for example, an insulator. The supporting member **12** supports one end of the spring structure **11**. The one end of the spring structure **11** penetrates the supporting member **12**. Further, the supporting member **12** is fixed to the upper surface of the board **41** in such a way as to cover one end of the wiring pattern **42**. Then, the one end of the spring structure **11**, the one end penetrating the supporting member **12**, and the one end of the wiring pattern **42**, the one end being covered by the supporting member **12**, are connected.

The supporting member **13** constitutes an other end side supporting member. This supporting member **13** is formed of, for example, an insulator. The supporting member **13** supports the other end of the spring structure **11**. The other end of the spring structure **11** penetrates the supporting member **13**.

The plug **14** is provided for the support member **13**. This plug **14** has a connection portion **14a** in the inside thereof. This connection portion **14a** is connected to the other end of the spring structure **11**, the other end penetrating the support member **13**. On the other hand, the socket **15** is provided for the one end of the electric wire **51**. This socket **15** has a connection portion **15a**. This connection portion **15a** is provided in a receptacle of the socket **15**. Then, the plug **14** can be inserted into the receptacle of the socket **15**. Further, the connection portion **14a** of the plug **14** and the connection portion **15a** of the socket **15** can be in contact with each other.

More specifically, when the plug **14** is inserted into the inside of the socket **15**, the connection portion **14a** of the plug **14** and the connection portion **15a** of the socket **15** are brought into contact with each other, as shown in FIGS. 2 and 3. In contrast with this, when the plug **14** is withdrawn from the inside of the socket **15**, the spring structure **11** enters a state in which the spring structure is mechanically and electrically connected to the board **41** while enters a

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state in which the spring structure is not mechanically and electrically connected to the electric wire **51**.

At the time of insertion of the plug **14** into the socket **15**, when a vibration occurs in the housing **43** and the vibration propagates from the housing **43**, via the spring structure **11**, toward the electric wire **51**, the vibration is attenuated by the elastic deformation of the spring structure **11**. On the other hand, at the time of insertion of the plug **14** into the socket **15**, when a vibration occurs in the electric wire **51** and the vibration propagates from the electric wire **51**, via the spring structure **11**, toward the housing **43**, the vibration is attenuated by the elastic deformation of the spring structure **11**.

Further, at the time of insertion of the plug **14** into the socket **15**, when a noise occurs in the wiring pattern **42** of the board **41** and the noise propagates from the wiring pattern **42**, via the spring structure **11**, toward the signal line **51a** of the electric wire **51**, the noise is removed by the inductance component of the spring structure **11**. On the other hand, when a noise occurs in the signal line **51a** of the electric wire **51** and the noise propagates from the signal line **51a**, via the spring structure **11**, toward the wiring pattern **42**, the noise is removed by the inductance component of the spring structure **11**.

As mentioned above, the connector **10** according to Embodiment 1 includes: the spring structure **11** having a spring property as a result of being constituted by a conducting wire wound spirally, and functioning as a coil having an inductance component; the support member **12** on a one end side to support the one end of the spring structure **11**, and to connect the one end of the spring structure **11** to the wiring pattern **42** provided on the board **41**; and the support member **13** on an other end side to support the other end of the spring structure **11**, and to electrically connect the other end of the spring structure **11** to the electric wire **51**. Therefore, the connector **10** can achieve both the absorption of vibrations and the removal of noises using the single spring structure **11**.

Further, the connector **10** includes the plug **14** provided for the support member **13** and connected to the other end of the spring structure **11**, and the socket **15** which is provided for the electric wire **51** and into which the plug **14** can be inserted. Therefore, the connector **10** makes it easy to make a mechanical connection and an electrical connection to the electric wire **51**.

Embodiment 2

A connector **20** according to Embodiment 2 will be explained using FIGS. 4 to 6. Components having the same functions as those of components explained in the connector **10** according to Embodiment 1 are denoted by the same reference signs, and an explanation of the components will be omitted.

FIG. 4 is a plane view showing the configuration of the connector **20** according to Embodiment 2. FIG. 5 is a side view showing the configuration of the connector **20** according to Embodiment 2. FIG. 6A is an outline view showing a double spring structure **21** which is viewed from an axis direction thereof. FIG. 6B is an outline view showing the double spring structure **21** which is viewed from the outside in a radial direction thereof.

As shown in FIGS. 4 and 5, the connector **20** according to Embodiment 2 electrically connects a single board **41** and a single cable **61** by means of a pair of transmission lines, for example. One end of the connector **20** is mechanically

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and electrically connected to the board 41, and the other end of the connector 20 is mechanically and electrically connected to the cable 61.

The board 41 has a pair of wiring patterns 42. These wiring patterns 42 are arranged in parallel.

The cable 61 is the one in which a pair of electric wires 51 are covered by a sheath 61a, for example. The sheath 61a is formed of, for example, a resin material. At one end of the cable 61 is disposed a socket 25.

The connector 20 includes the spring structure 21, support members 22 and 23, a plug 24 and the socket 25.

The spring structure 21 is supported between the support members 22 and 23. This spring structure 21 has a function of absorbing mechanical vibrations and a function of removing electrical noises.

Concretely, as shown in FIGS. 6A and 6B, the spring structure 21 is the one in which two spring structures 11 are superposed on each other. In other words, the spring structure 21 is constituted by a pair of conducting wires wound spirally and alternately each other. Thus, the pair of conducting wires which form a double helix structure are provided coaxially and in the same winding direction, and each of the outer peripheral surfaces of the pair of conducting wires is covered with an insulator (not illustrated).

Therefore, the spring structure 21 has a spring property. More specifically, the spring structure 21 can be elastically deformed in an axis direction thereof. The spring structure 21 also functions as a coil having an inductance component. In addition, because the outer peripheral surface of each of the conducting wires is covered with an insulator, the spring structure 21 is placed in a state in which the conducting wires do not conduct with each other even though the conducting wires come into contact with each other.

The insulator provided on the outer peripheral surface of each of the conducting wires has elasticity. Therefore, even though the pair of conducting wires in the spring structure 21 are elastically deformed, the insulator can follow the elastic deformation.

The support member 22 constitutes a one end side support member. This support member 22 is formed of, for example, an insulator. The support member 22 supports each of one ends of the pair of conducting wires in the spring structure 21. More specifically, the support member 22 supports one end of each of the spring structures 11. Each of the one ends of the pair of conducting wires in the spring structure 21 penetrates the support member 22.

Further, the support member 22 is fixed to an upper surface of the board 41 in such a way as to cover each of one ends of the pair of wiring patterns 42. Then, the one end of one of the conducting wires, the one end penetrating the support member 22, and the one end of one of the wiring patterns 42, the one end being covered by the support member 22, are connected. On the other hand, the one end of the other conducting wire, the one end penetrating the support member 22, and the one end of the other wiring pattern 42, the one end being covered by the support member 22, are connected.

The support member 23 constitutes an other end side support member. This support member 23 is formed of, for example, an insulator. The support member 23 supports each of other ends of the pair of conducting wires in the spring structure 21. More specifically, the support member 23 supports the other end of each of the spring structures 11. Each of the other ends of the pair of conducting wires in the spring structure 21 penetrates the support member 23.

The plug 24 is provided for the support member 23. This plug 24 has a pair of connection portions 24a in the inside

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thereof. One end of one of the connection portions 24a is connected to the other end of one of the conducting wires, the other end penetrating the support member 23. Further, one end of the other connection portion 24a is connected to the other end of the other conducting wire, the other end penetrating the support member 23.

On the other hand, the socket 25 is provided for the one end of the cable 61. This socket 25 has a pair of connection portions 25a. The pair of connection portions 25a are provided in the inside of a receptacle of the socket 25.

The plug 24 can be inserted into the receptacle of the socket 25. Further, the one of the connection portions 24a and one of the connection portions 25a can be in contact with each other. On the other hand, the other connection portion 24a and the other connection portion 25a can be in contact with each other.

More specifically, when the plug 24 is inserted into the inside of the socket 25, the pair of connection portions 24a in the plug 24 are in contact with the pair of respective connection portions 25a in the socket 25. In contrast with this, when the plug 24 is withdrawn from the inside of the socket 25, the spring structure 21 enters a state in which the spring structure is mechanically and electrically connected to the board 41 while enters a state in which the spring structure is not mechanically and electrically connected to the cable 61.

At the time of insertion of the plug 24 into the socket 25, when a vibration occurs in a housing 43 and the vibration propagates from the housing 43, via the spring structure 21, toward the cable 61, the vibration is attenuated by the elastic deformation of the spring structure 21. On the other hand, at the time of insertion of the plug 24 into the socket 25, when a vibration occurs in the cable 61 and the vibration propagates from the cable 61, via the spring structure 21, toward the housing 43, the vibration is attenuated by the elastic deformation of the pair of conducting wires in the spring structure 21.

Here, a pair of transmission lines are formed between the board 41 and the cable 61, as can be seen clearly from the above-mentioned explanation. One of the transmission lines is constituted by one of the electric wires 51 in the cable 61, the one of the conducting wires in the spring structure 21 and the one of the wiring patterns 42 in the board 41. The other transmission line is constituted by the other electric wire 51 in the cable 61, the other conducting wire in the spring structure 21 and the other wiring pattern 42 in the board 41.

When currents flow through the pair of conducting wires in the spring structure 21 in the same direction, the conducting wires mutually strengthen their respective magnetic fields caused by the currents flowing through the conducting wires. Therefore, a noise occurring from the board 41 or the cable 61 is removed by the inductance component of each of the conducting wires. The above-mentioned noise is referred to as a so-called common mode noise.

As mentioned above, in the connector 20 according to Embodiment 2, the spring structure 21 is constituted by the pair of conducting wires wound spirally and alternately each other. The one ends of the pair of conducting wires are connected to the pair of respective wiring patterns 42 provided on the board 41. The other ends of the pair of conducting wires are connected to the pair of respective electric wires 51 contained in the single cable 61. The pair of conducting wires are provided coaxially and in the same winding direction, and an insulator is provided on each of the outer peripheral surfaces of the pair of conducting wires. Therefore, the connector 20 can achieve both the absorption

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of vibrations and the removal of common mode noises using the single spring structure 21.

Further, in the connector 20, the insulator provided on each of the outer peripheral surfaces of the pair of conducting wires in the spring structure 21 has elasticity. Therefore, even though the pair of conducting wires in the spring structure 21 be elastically deformed, the insulator can follow the elastic deformation.

Embodiment 3

A connector 30 according to Embodiment 3 will be explained using FIG. 7. Components having the same functions as those of components explained in the connector 20 according to Embodiment 2 are denoted by the same reference signs, and an explanation of the components will be omitted.

FIG. 7 is a plane view showing the configuration of the connector 30 according to Embodiment 3.

As shown in FIG. 7, the connector 30 according to Embodiment 3 electrically connects a single board 41 and multiple cables 61 by means of the same number of transmission lines as the number of cables 61, for example. One end of the connector 30 is mechanically and electrically connected to the board 41, and the other end of the connector 30 is mechanically and electrically connected to the cables 61. FIG. 7 shows an example in which the number of cables 61 is two.

When one pair of wiring patterns 42 are defined as one set, the board 41 has two sets of wiring patterns. These wiring patterns 42 are arranged in parallel. Further, to one ends of the two cables 61 is provided a single socket 35.

The connector 30 includes two spring structures 21, a single support member 32, two support members 23, a single plug 34, and the single socket 35.

The two spring structures 21 are arranged in parallel. These spring structures 21 are supported between the support member 32 and the support members 23. The number of installed spring structures 21 is the same as the number of sets of pair of wiring patterns 42 and the number of cables 61.

The support member 32 constitutes a one end side support member. This support member 32 is formed of, for example, an insulator. The support member 32 supports each of one ends of a pair of conducting wires in each of the spring structures 21. Each of the one ends of the pair of conducting wires in each of the spring structures 21 penetrates the support member 32.

Further, the support member 32 is fixed to an upper surface of the board 41 in such a way as to cover each of one ends of the two sets of wiring patterns 42. Then, one end of one of the conducting wires in each of the pairs, the one end penetrating the support member 32, and one end of one of the wiring patterns 42 in each of the sets, the one end being covered by the support member 32, are connected. On the other hand, one end of the other conducting wire in each of the pairs, the one end penetrating the support member 32, and one end of the other wiring pattern 42 in each of the sets, the one end being covered by the support member 32, are connected.

One of the support members 23 supports each of the other ends of the pair of conducting wires in one of the spring structures 21. The other support member 23 supports each of the other ends of the pair of conducting wires in the other spring structure 21. Each of the other ends of the pair of conducting wires in the one of the spring structures 21 penetrates the one of the support members 23. Each of the

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other ends of the pair of conducting wires in the other spring structure 21 penetrates the other support member 23.

The plug 34 is provided in such a way as to cross over the two support members 23. When one pair of connection portions 34a are defined as one set, this plug 34 has two sets of connection portions in the inside thereof. Each of one ends of one of the sets of connection portions 34a is connected to the other ends of the pair of conducting wires, the other ends penetrating the one of the support members 23. Each of one ends of the other set of connection portions 34a is connected to the other ends of the pair of conducting wires, the other ends penetrating the other support member 23.

In contrast with this, the socket 35 is provided in such a way as to cross over the one ends of the two cables 61. When one pair of connection portions 35a are defined as one set, this socket 35 has two sets of connection portions. The connection portions 35a in each of the sets is provided in a receptacle of the socket 35.

The plug 34 can be inserted into the receptacle of the socket 35. Further, the one of the sets of connection portions 34a and the one of the sets of respective connection portions 35a can be in contact with each other. On the other hand, the other set of connection portions 34a and the other set of respective connection portions 35a can be in contact with each other.

More specifically, when the plug 34 is inserted into the inside of the socket 35, the two sets of connection portions 34a in the plug 34 are connected to the two sets of respective connection portions 35a in the socket 35. In contrast with this, when the plug 34 is withdrawn from the inside of the socket 35, the two spring structures 21 enter a state in which the spring structures are mechanically and electrically connected to the single board 41 while enters a state in which the spring structures are not mechanically and electrically connected to the two cables 61.

As mentioned above, in the connector 30 according to Embodiment 3, the same number of spring structures 21 each of which is constituted by a pair of conducting wires wound spirally and alternately each other are provided as the number of sets of wiring patterns 42 when one pair of wiring patterns are defined as one set, and as the number of cables 61. Therefore, the connector 30 can achieve both the absorption of vibrations and the removal of common mode noises even using the multiple spring structures 21.

It is to be understood that an arbitrary combination of two or more of the above-mentioned embodiments can be made, various changes can be made in an arbitrary component in each of the above-mentioned embodiments, or an arbitrary component in each of the above-mentioned embodiments can be omitted within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

The connector according to the present disclosure can achieve both the absorption of vibrations and the removal of noises by including a spring structure constituted by a conducting wire wound spirally, and thus is suitable for use as a connector and so on.

REFERENCE SIGNS LIST

10, 20, 30 connector, 11, 21 spring structure, 12, 13, 22, 23, 32 support member, 14, 24, 34 plug, 15, 25, 35 socket, 14a, 15a, 24a, 25a, 34a, 35a connection por-

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tion, **41** board, **42** wiring pattern, **43** housing, **51** electric wire, **51a** signal line, **51b** insulator, **61** cable, and **61a** sheath.

The invention claimed is:

1. A connector comprising:

a first spring structure having:

a first conducting wire wound spirally and having a spring property and an inductance component;

a second conducting wire wound spirally and having a spring property and an inductance component, the second conducting wire being wound alternately and coaxially in a same winding direction with respect to the first conducting wire;

a first support member supporting a first end of the first conducting wire and a first end of the second conducting wire, and configured to connect the first end of the first conducting wire and the first end of the second conducting wire to separate wiring patterns provided on a board; and

a second support member supporting a second end of the first conducting wire, opposite to the first end of the first conducting wire, and a second end of the second conducting wire opposite to the first end of the second conducting wire, to electrically connect the second end of the first conducting wire and the second end of the second conducting wire to separate electric wires contained in a single cable,

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wherein an insulator is provided on an outer peripheral surface of each of the first and second conducting wires.

2. The connector according to claim **1**, wherein the insulator has elasticity.

3. The connector according to claim **1**, further comprising:

a second spring structure including:

a third conducting wire wound spirally and having a spring property and an inductance component; and

a fourth conducting wire wound spirally and having a spring property and an inductance component, the fourth conducting wire being wound alternately and coaxially in a same winding direction with respect to the third conducting wire;

wherein the wiring patterns and the single cable are provided for each of the first spring structure and the second spring structure.

4. The connector according to claim **1**, further comprising:

a plug provided for the second support member and connected to the second end of the first conducting wire and the second end of the second conducting wire; and a socket which is provided for the separate electric wires and into which the plug can be inserted.

* * * * *